TERRAFORM WORKSPACE AND LOOPS

TERRAFORM WORKSPACE

A workspace is an isolated environment with a separate state file within the same Terraform configuration.

ENVIRONMENTS LIKE

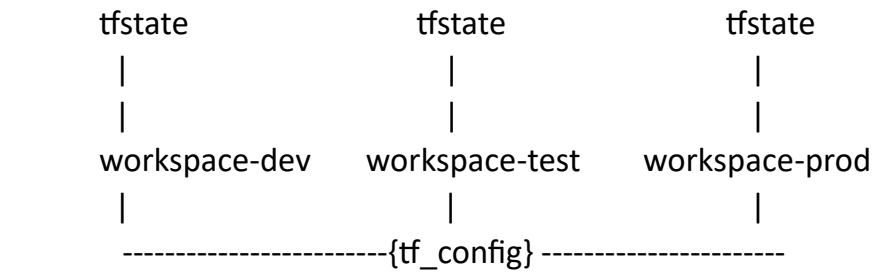
1. default → production
2. dev → development
3. qa → testing

Each workspace keeps its own state file, so resources don't clash.

COMMANDS

1. terraform workspace new dev --------> To create a dev workspace
2. terraform workspace list ----------> to list the workspace
3. terraform workspace show ----------> to show the workspace
4. terraform workspace select dev -----> to select the dev workspace
5. terraform workspace delete dev -----> to delete the workspace

WORKING



ON DEFAULT_WORKSPACE

1.main.tf

```
provider "aws" {
  region  = "ap-south-1"
  profile = "config"
}

resource "random_id" "random_id" {
  byte_length = 8
}
resource "aws_s3_bucket" "demo-bucket" {
```

```
  bucket = "demo-bucket-${terraform.workspace}-${random_id.random_id.hex}"
}
```

1. terraform init
2. terraform apply –auto-approve

check on the s3, bucket is created by default workspace
{demo-bucket-default-a1b2c3d4e5f6g7h8}

ON WORKSPACE_DEV

1. terraform workspace new dev
2. terraform workspace show
3. Terraform init
4. Terraform apply –auto-approve

5. check it on the s3
    1. demo-bucket-dev-98ff12ca4433aabb
    2. demo-bucket-default-a1b2c3d4e5f6g7h8

LOOPS

1. FOR EACH

1. for_each is used to create multiple resources using a single resource block.
2. It is useful when resources need different names, values, or configurations.
3. Each resource is created using a unique key from a map or set.
4. It reduces code duplication and improves readability.

1.main.tf

```
resource "aws_instance" "my_instance" {
  for_each = tomap({
    server-1 = "t3.micro",
    server-2 = "t3.small"
  })
  ami                = “ami-0685bcc683dadb6b9”
  instance_type = each.value
  tags      = {
    Name = each.key
  }
}
```

2.output.tf

```
output "aws_instance_public_ip" {
  value = [for instance in aws_instance.my_instance : instance.public_ip]
}
output "aws_instance_private_ip" {
  value = [for instance in aws_instance.my_instance : instance.private_ip]
}
```

1. terraform init
2. terraform apply –auto-approve
3. terraform destroy –auto-approve

2.COUNT

1.provider.tf

```
provider "aws" {
   region = "ap-south-1"
   profile = "config"
}
```

2.main.tf

```
resource "aws_instance" "my_instance" {
   ami = "ami-0685bcc683dadb6b9"
   instance_type = "t3.micro"
   count = 2
   tags = {
      Name = "my-instance"
   }
}
```

1. git add .
2. git commit -m “count”
3. git push

ON UBUNTU-LOCAL

1. git clone https://github.com/mayurpandit25/terraform-b30.git
2. cd terraform-b30
3. terraform init
4. terraform apply –auto-approve
5. terraform destroy –auto-approve